

Problema 12 [Ref. 162 — País Basc, EAU Juny 2019, Exercici A3]

Siguin A i B dos successos tals que:

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{3}, \quad P(A \cup B) = \frac{3}{4}.$$

Calculeu:

- a) $P(A | B)$.
- b) La probabilitat que **no ocorri cap** dels dos successos.
- c) La probabilitat que ocorri A i **no** ocorri B .
- d) La probabilitat que ocorri **exactament un** dels dos successos.

$$P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{3}{4} = \frac{6+4-9}{12} = \frac{1}{12}.$$

$$\text{a) } P(A|B) = \frac{1/12}{1/3} = \frac{1}{4}.$$

$$\text{b) } P(A^c \cap B^c) = 1 - \frac{3}{4} = \frac{1}{4}.$$

$$\text{c) } P(A \cap B^c) = \frac{1}{2} - \frac{1}{12} = \frac{5}{12}.$$

$$\text{d) } P(\text{exactament un}) = \frac{3}{4} - \frac{1}{12} = \frac{2}{3}.$$